Model	Long Sequence			Reasoning		
	Throughput ↑	Avg. JCT ↓	Padding Ratio ↓	Throughput ↑	Avg. JCT ↓	Padding Ratio ↓
EGTP (Ours)	**131.05**	**4.20**	**0.18**	**194.21**	**8.21**	**0.09**
SSJF-Reg	117.87	11.52	0.57	146.74	9.32	0.33
SSJF-MC	109.50	23.24	0.38	124.34	22.31	0.31
S3	115.09	15.74	0.55	139.89	10.31	0.42
PiA	119.20	18.40	0.47	141.83	16.74	0.31
TPV	116.03	43.10	0.57	142.41	40.12	0.32
TRAIL	**129.58**	**9.45**	0.51	147.03	9.32	0.34
LTR-C	127.14	12.01	**0.21**	**150.57**	**9.30**	**0.14**

Table 2: End-to-End System Performance Comparison on Different Scenarios (**Best**, **Second Best**)

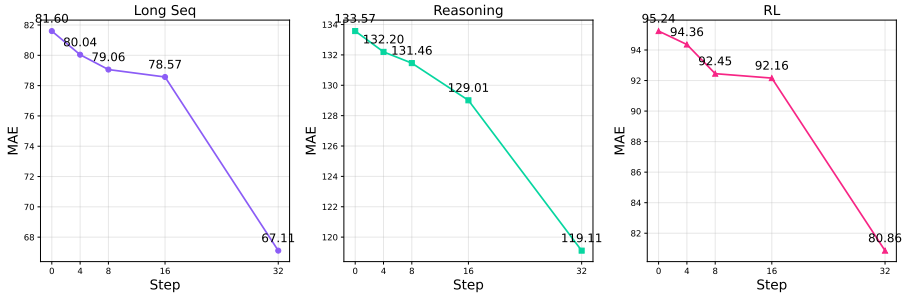


Figure 3: MAE improvements for Progressive Length Prediction.

in Table 3, clearly demonstrates that our proposed EGTP method outperforms the baselines across all tasks. EGTP achieves an average MAE of 103.47, which is a significant improvement over the best-performing baseline, Max Pooling, at 119.69. The advantage of EGTP is especially prominent on the Long Sequence task, where its MAE of 81.60 is substantially lower than any competing method. This result confirms the superiority of our approach in effectively capturing key features from complex sequences, a task where traditional pooling strategies tend to fall short.

The sensitivity analysis of the hyperparameter λ and its effect on the joint optimization is discussed in detail in Appendix D. Additionally, detailed experimental results for the Qwen2.5-0.5B and Qwen2.5-1.5B models can be found in Appendix E.5. And the length prediction performance comparison on various other datasets is provided in the Appendix E.6.

Table 3: Ablation study on the impact of different pooling methods.

Pooling Method	Reasoning	Long Sequence	RL	Average
EGTP (Ours)	**133.57**	**81.60**	**95.24**	**103.47**
Average Pooling	142.40	173.85	149.92	155.39
Max Pooling	137.88	122.74	98.46	119.69
Last Token Pooling	139.09	135.44	105.39	126.64

5 CONCLUSION

In this paper, we propose a novel framework that predicts sequence length by reusing the model's own internal activations. This approach circumvents the overhead and generalization failures of separate, auxiliary predictors. Our method introduces EGTP for static estimation and PLP for progressive prediction in dynamic environments. We validate our approach on ForeLen, a new and challenging benchmark we developed for this task. The results demonstrate superior prediction accuracy, confirming that sufficient signals for length determination are already encoded within the LLM's hidden states.

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REPRODUCIBILITY STATEMENT

To facilitate reproducibility, we provide detailed descriptions of the experimental setup and hyperparameter configurations in Section 4.2 The full implementation of our method, is publicly available at: `https://github.com/xiehuanyi/LP_Bench`. The dataset used in our experiments is publicly accessible at: `https://huggingface.co/datasets/abinzzz/ForeLen`.

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